Junyao Dong

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Education

Nanyang Technological University — Singapore
M.Eng. by Research in Computing and Data Science Engineering — 08/2024 – 06/2026
- GPA: 4.17/5.0. Co-advised by Cheng Long and Shuhao Zhang.
- **Research Interests:** Vector databases, retrieval systems, and machine learning systems.

The University of Manchester — Manchester, UK
Visiting Student in Computer Science — 08/2019 – 05/2020
- Overall Average: 70% (First-Class Standard).

Guangdong University of Foreign Studies — Guangzhou, China
B.Eng. in Computer Science — 09/2017 – 06/2021
- GPA: 3.4/4.0.

Publications (* = Equal contribution)

The Program Counter Is a Path: Code, Data, and Control in One Addressable Tree
Peng Li*, Junyao Dong*
- Submitted to *Proceedings of the ACM on Programming Languages*, 2027 (**PLDI'27**). [Website] [Code]

Shared Memory in Multi-Agent Systems: Which View Does the Agent Read?
Junyao Dong, Xisen Wang, Yu Chen, Yi Li, Qi Zhang
- Submitted to the *26th International Conference on Autonomous Agents and Multiagent Systems* (**AAMAS'27**).

ANNchor: Concurrency Control for Freshness in Streaming Graph-Based Approximate Nearest Neighbor Search
Junyao Dong, Shuhao Zhang, Wenchao Zhou, Cheng Long
- Revision at *Proceedings of the ACM on Management of Data*, 2027 (**SIGMOD'27**, Round 3; decision Dec 12). [Latest Status]

CANDOR-Bench: Benchmarking In-Memory Continuous ANNS under Dynamic Open-World Streams
Mingqi Wang*, Junyao Dong*, Zhuoyan Wu, Jun Liu, Ruicheng Zhang, Jianjun Zhao, Ruipeng Wan, Xinyan Lei, Shuhao Zhang, Bolong Zheng, Haikun Liu, Xiaofei Liao, Hai Jin
- *Proceedings of the ACM on Management of Data*, 2026 (**SIGMOD'26**). [Paper] [Code]

Research Experience († = Full-time)

Vector Database Group @ NTU — Singapore
Research Associate† (Advised by Cheng Long) — 08/2026 – Present

Applying vector search and RaBitQ quantization to long-context LLM inference.
- Studying retrieval over the KV cache for sparse attention in long-context decoding.

PageIndex — Remote
Research Intern (Co-advised by Mingtian Zhang and Yu Tang) — 12/2025 – Present

Building retrieval over long, hierarchically structured documents.
- Led the design and implementation of ConDB, a chunk-free, vector-free context database for long documents.
- Collaborated with MongoDB on storing and searching large hierarchical JSON in the MongoDB Python driver.

Vector Database Group @ NTU & IntelliStream @ HUST — Singapore
Graduate Student Researcher† (Co-advised by Cheng Long and Shuhao Zhang) — 08/2024 – 06/2026

Investigated index freshness in streaming vector search, and how to benchmark it.
- Designed ANNchor, a concurrency-control framework that guarantees post-acknowledgment visibility on a single live HNSW graph; up to 4.6× median search throughput over global locking and 68% lower median end-to-end P99 latency under write-heavy load.
- Co-developed CANDOR-Bench, a continuous ANNS benchmark of 19 algorithms on 12 datasets under drift and concurrent read–write streams; co-designed the streaming evaluation harness.

The Hong Kong University of Science and Technology (Guangzhou) Guangzhou, China
Research Assistant† (Advised by Ying Cui) 04/2024 – 06/2024

Implemented adaptive video delivery for variable network conditions.

- Developed an adaptive bitrate system that cuts playback buffering without losing visual quality.

Selected Open-Source Projects

RaBitQ-Library **github.com/VectorDB-NTU/RaBitQ-Library** ∼300 GitHub stars
Contributor

Official library for RaBitQ, a randomized bitwise quantization method for approximate nearest neighbor search, covering quantization, scanning, and index integration (used by Meta, NVIDIA, Microsoft, and Alibaba).

- Hardened IVF index construction robustness and prototyped quantized graph-index search integration.

RL-Kernel **github.com/RL-Align/RL-Kernel** 400+ GitHub stars
Contributor

High-performance RL post-training infrastructure for operator-level train–inference consistency.

- Implemented CUDA/Triton kernels for DeepSeek V4 compressed attention: one-softmax MQA with an attention sink, and grouped output projection.

vllm-rlt **github.com/InferMatrixLab/vllm-rlt** 200+ GitHub stars
Contributor

Inference engine for looped language models. Recurrent steps are batched continuously, so requests at different loop depths share one batch.

- Designed the engine loop so synchronous and asynchronous steps share scheduling and result handling, while submission, GPU completion, and output delivery stay separate.

Industry Experience († = Full-time)

Baidu Shanghai, China
Software Engineer†, Cloud Services 05/2021 – 03/2024

Built live-video infrastructure, from edge ingest through storage to on-air production.

- Integrated H.264/H.265 at the edge for Baidu Cloud Live Streaming Service and wrote outputs as MP4, FLV, and HLS.
- Added watermarks, picture-in-picture, transitions, and subtitles to Baidu Cloud Live Cast Production Studio on a customized FFmpeg engine and Baidu's RTC stack.

Momenta Suzhou, China
Research and Development Intern, Autonomous Driving 10/2020 – 05/2021

Developed log infrastructure for replaying recorded drives as training data.

- Synchronized perception and planning streams onto one timeline.

Awards and Honors

Second Prize, National Modeling Competition 2020

Overseas Student Programme Award 2020

Academic Service

Availability and Reproducibility Committee (ARC)

- SIGMOD 2026 2026

External Reviewer

- VLDB 2026, SIGMOD 2027, VLDB 2027, WSDM 2027 2026
- ICDE 2026 2025